FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2021

TEST DATE: 09-05-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. At what altitude will the acceleration due to gravity be 25% of that at the earth's surface (given radius of earth is R):
   (A) $R/4$
   (B) R
   (C) $3R/8$
   (D) $R/2$

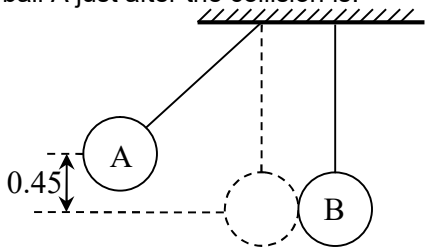
2. A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre:
   (A) Increases
   (B) Decreases
   (C) Remains same
   (D) During the compression increases then returns at the previous value.

3. The angular velocity and the amplitude of a simple pendulum are ω and A respectively. At a displacement x from the mean position if its kinetic energy is T and potential energy is U, then the ratio of T to U is:
   (A) $\dfrac{x^2\omega^2}{\left(A^2 - x^2\omega^2\right)}$
   (B) $\dfrac{x^2}{\left(A^2 - x^2\right)}$
   (C) $\dfrac{A^2 - x^2\omega^2}{\left(x^2 - x^2\omega^2\right)}$
   (D) $\dfrac{A^2 - x^2}{x^2}$

4. A gas is filled in a cylinder, its temperature is increased by 20% on Kelvin scale and volume if reduced by 10%. How much percentage of the gas will leak out?
   (A) 30%
   (B) 40%
   (C) 15%
   (D) 25%

5. A boat of length 10m and mass 450 kg is floating without motion in still water. A man of mass 50 kg standing at one end of it, walks to the other end of it and stops. The magnitude of the displacement of the boat in meter relatively to ground is:
   (A) Zero
   (B) 1m
   (C) 2m
   (D) 5m

6. Two massless string of length 5m hang from the ceiling very near to each other as shown in the figure. Two balls A and B of masses 0.25 kg and 0.5 kg are attached to string. The ball A is released from rest at a height 0.45 m as shown in the figure. The collision between two balls is completely elastic. Immediately after the collision, the kinetic energy of ball B is 1J. The velocity of ball A just after the collision is:



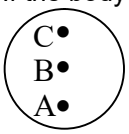
(A) $5\,ms^{-1}$ to the right

(B) $5\,ms^{-1}$ to the left

(C) $1\,ms^{-1}$ to the right

(D) $1\,ms^{-1}$ to the left

7. If the body rolls perfectly on a surface then:



(A) $V_A = V_B = V_C$

(B) $V_A > V_B > V_C$

(C) $V_C > V_B > V_A$

(D) None of these

8. A steel scale measures the length of a copper wires as 80.0 cm, when both are at 20°C (the calibration temperature for scale). What would be the scale read for the length of the copper wire when both are at 40°C?

(Given $\alpha_{steel} = 11\times10^{-6}\ per°C$ and $\alpha_{copper} = 17\times10^{-6}\ per°C$)

(A) $80.0096\,\text{cm}$

(B) $80.0272\,\text{cm}$

(C) $1\,\text{cm}$

(D) $25.2\,\text{cm}$

9. A tuning fork A of frequency 512Hz produces 5 beats per second when sounded with tuning fork B. If B is loaded with wax, the number of beats is again 5 per second. What is the frequency of tuning fork B which not loaded?

(A) $517\,\text{Hz}$

(B) $507\,\text{Hz}$

(C) $510\,\text{Hz}$

(D) $512\,\text{Hz}$

10. A reflecting telescope is to be made by using spherical mirror of radius of curvature 1.2 m and an eye piece with a focal length of 1.25 cm. For maximum strain position of eye what will be the angular magnification? (For normal eye least distance of vision is 25 cm):
(A) 22.7
(B) 50.4
(C) 48.5
(D) 118.2

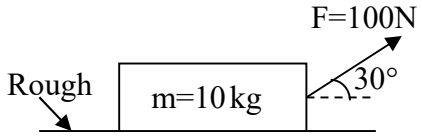
11. Three bodies, a ring, a solid cylinder and solid sphere rolls down the same inclined plane without slipping. They start from rest and their radius are identical. Then the body which reaches to the ground with maximum velocity is:
(A) Sphere
(B) Ring
(C) Cylinder
(D) All reaches with same speed

12. The electric potential in a certain region of space depends only on x coordinate as $V = -\alpha x^3 + \beta$, here α and β are constant. The volume charge distribution of space charge is given as:
(A) $\rho(x) = 3\alpha\varepsilon_0 x^2$
(B) $\rho(x) = -3\alpha\varepsilon_0 x^2$
(C) $\rho(x) = 6\alpha\varepsilon_0 x^2$
(D) $\rho(x) = 6\alpha\varepsilon_0 x$

13. Two plane mirror are inclined at an angle of 60° with each other. A ray of light travelling horizontally is reflected first from one mirror and then from the other. The resultant deviation is:
(A) 60°
(B) 120°
(C) 180°
(D) 240°

14. A particle is projected at t = 0 with velocity u at angle θ with the horizontal. Then the ratio of the tangential acceleration and the radius of curvature at the point of projection is:
(A) $\frac{g^2 \sin 2\theta}{u^2}$
(B) $\frac{2g^2 \sin\theta}{u^2}$
(C) $\frac{g^2 \sin\theta}{u^2}$
(D) $\frac{g^2 \sin 2\theta}{2u^2}$

15. A block of mass 10 kg is placed on a horizontal ground surface as shown in the figure and a force F = 100 N is applied on the block as shown in the figure. The block is at rest with respect to ground. If the contact force between block and ground is 25n (in Newton) then value of n is: (Take g = 10 m/s^2)



(A) 1
(B) 2
(C) 3
(D) 4

16. A rod of length ℓ lies on the x-axis with its left end at the origin. An external force pulls the rod in y-direction with a force $F = R_0\left(\frac{\ell - x}{\ell}\right)$. If the axis is fixed at $x = 0$ then the torque is:

(A) $-F_0\ell$
(B) $\frac{F_0\ell}{2}$
(C) $\frac{-F_0\ell}{2}$
(D) None of these

17. A particle in a certain conservative force field has a potential energy given by $U=\frac{20yz}{x}$. The force exerted on it is:

(A) $\left(\frac{20yz}{x^2}\right)\hat{i}+\left(\frac{20z}{x}\right)\hat{j}+\left(\frac{20y}{x}\right)\hat{k}$
(B) $\left(\frac{20yz}{x^2}\right)\hat{i}-\left(\frac{20z}{x}\right)\hat{j}-\left(\frac{20y}{x}\right)\hat{k}$
(C) $-\left(\frac{20yz}{x^2}\right)\hat{i}-\left(\frac{20z}{x}\right)\hat{j}-\left(\frac{20y}{x}\right)\hat{k}$
(D) $-\left(\frac{20yz}{x^2}\right)\hat{i}+\left(\frac{20z}{x}\right)\hat{j}+\left(\frac{20y}{x}\right)\hat{k}$

18. Through an ideal inductor coil of L = 0.2 H, an AC current of amplitude 2A is passed, first with frequency 'f_1' then with frequency'f_2'. Then ratio of maximum value of induced emf in two cases e_1 and e_2 respectively. Find $\left(\frac{e_1}{e_2}\right)$ in the coil, in the two cases is:

(A) f_1 / f_2
(B) f_2 / f_1
(C) $\left(f_1 / f_2\right)^2$
(D) $1:1$

19. Calculate the conductivity of pure silicon at room temperature when concentration of carrier is 1.6×10^{10} per-cm^3.

Take $\mu_e = 1500\,cm^2/volt\text{-}sec, and\ \mu_h = 500\,cm^2/volt\text{-}sec\ at\ room\ temp.$

(A) 5.12×10^{-6} moh/cm

(B) 3.12×10^{-6} moh/cm

(C) 2.35×10^{-6} moh/cm

(D) None of these

20. The rms value of the electric field of the light coming from the sun is 720 N/C. The average total energy density of the electromagnetic wave is:

(A) $4.58\times10^{-6}\ J/m^3$

(B) $6.37\times10^{-9}\ J/m^3$

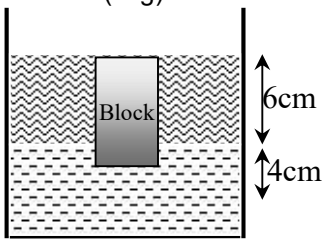
(C) $81.35\times10^{-12}\ J/m^3$

(D) $3.3\times10^{-3}\ J/m^3$

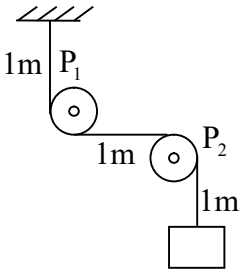
SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A cubical block of wood 10 cm on a side floats at the interface between oil and water with its lower surface horizontal and 4 cm below the interface. The density of oil is 0.6 gcm^{-3}. The mass of block (in g) is:



22. A copper wire of length 3m and area of cross-section 1 mm^2, passes through an arrangement of two frictionless pulleys, P_1 and P_2. One end of the wire is rigidly clamped and a mass of 1kg is hanged from the other end. If the Young's modulus for copper is $10\times10^{10}\ N/m^2$, then the elongation in the wire (in mm) is:



23. The escape velocity from a planet is v_0. The escape velocity from a planet having twice the radius but same density is nv_0 then n is:

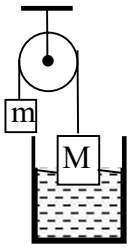
24. How long after the beginning of motion is the displacement of a harmonically oscillating point equal to one half its amplitude, if the period is 24 second and initial phase is zero?

25. A lead bullet at 27°C just melts when stooped by an obstacle. Assuming that 25% of heat is absorbed by the obstacle, then the velocity of the bullet at the time of striking:
(M.P. of lead = 327°C, specific heat of lead = 0.03 cal/g°C, latent heat of fusion of lead = 6 cal/g and J = 4.2 joule/cal) (in m/s):

26. Two wires are fixed in sonometer. Their tensions area in the ratio 8:1. The lengths are in the ratio 36:35. The diameters are in the ratio 4:1. Densities of the materials are in the ratio 1:2. If the higher frequency in the setting is 360 Hz, the beat frequency (per second) when the two wires sounded together is:

27. Two long straight wires A and B are placed 50 cm apart and carry current 20A and 15A respectively in same direction. A point P is 40cm from wire A and 30 cm from wire B. What is the magnitude of resultant magnetic field at 'P' in (μT) unit? $(\sqrt{2}=1.414)$

28. To one end of thread thrown over the pulley we suspended load of mass M = 10kg, made from a material specific gravity 2.5. The mass M is suspended in a container with water. By the other end of the thread we suspended mass m (see figure). For what minimum value of m (in kg) mass M can float in liquid in equilibrium? Friction is negligible everywhere.



29. A sphere of radius R have volume charge density give as

$$\rho(r) = Kr \; for \; r \le R$$

$$0 \; for \; r > R$$

If electric field at distance $\frac{R}{2}$ from centre is $\frac{KR^2}{N\varepsilon_0}$. Then N will be:

30. Two tuning forks having frequency 256 Hz (A) and 262 Hz (B), tuning fork A produces some beats per second with unknown tuning fork, same unknown tuning fork produce double beats per second from B tuning fork then the frequency of unknown tuning fork is:

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

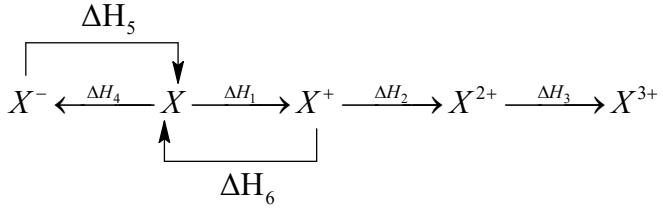
31. A fixed mass of an ideal gas contained in a 24.63L sealed rigid vessel at 1 atm is heated from $-73°C$ to $27°C$. Calculate change in Gibb's energy if entropy of gas is a function of temperature as $S=2+10^{-2}T\left(J/k\right)$: [Use 1atm L = 0.1 kJ]

(A) $1231.5J$
(B) $1281.5J$
(C) $781.5J$
(D) 0

32. The rate law for a reaction between the substances A and B is given by rate $= k\left[A\right]^{n}\left[B\right]^{m}$. On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be:

(A) $1/2^{m+n}$
(B) $2^{(m+n)}$
(C) $1/2^{n-m}$
(D) $2^{(n-m)}$

33. Consider the following sequence of reaction:



If electronic configuration of element X is $\left[Ne\right]3s^{1}$, then which of the following order is incorrect regarding given enthalpies?

(A) $\left|\Delta H_4\right|=\left|\Delta H_5\right|$
(B) $\left|\Delta H_2\right|>\left|\Delta H_1\right|$
(C) $\left|\Delta H_2\right|>\left|\Delta H_3\right|$
(D) $\left|\Delta H_1\right|=\left|\Delta H_6\right|$

34. The bond angle of Cl_2O is:

(A) Smaller than that of F_2O
(B) Greater than that of H_2O
(C) Smaller than that of H_2O
(D) Same as that of F_2O

35. For the reaction: $2NO(g)+Cl_2(g) \rightleftharpoons 2NOCl(g)$, NO and Cl_2 are initially taken in mole ratio $2:1$. The total pressure at equilibrium is found to be 1 atm. If mole of $NOCl$ are one fourth of that of Cl_2 at equilibrium, K_p for the reaction is:

(A) $13/36$
(B) $13/256$
(C) $13/512$
(D) $13/128$

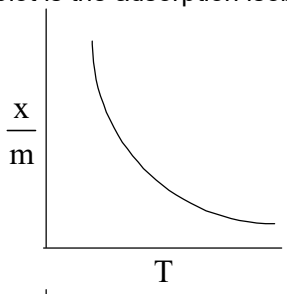
36. Suppose that in any Bohr atom or ion, orbits are only in even numbers like $2,4,6.....$ The maximum wavelength of radiation emitted in the visible region of hydrogen spectrum should be:

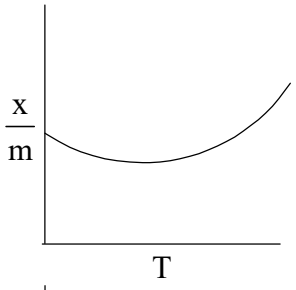
(A) $4/R$
(B) $R/4$
(C) $36/5R$
(D) $16/3R$

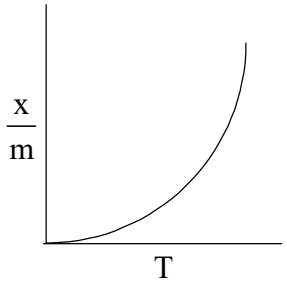
37. The overall charge present on the cyclic silicate anion $\left[Si_6O_{18}\right]^{n-}$ is:

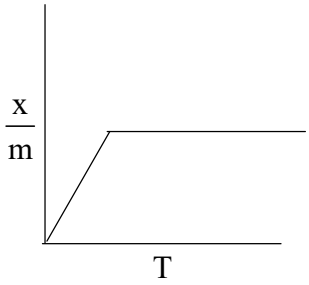
(A) 6
(B) 12
(C) 18
(D) 24

38. Which plot is the adsorption isobar for chemisorption?

(A)


(B)


(C)


(D)


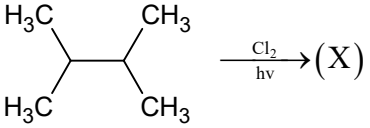
39. The solubility product of PbI_2 is 7.2×10^{-9}. The maximum mass of NaI which may be added in $500ml$ of $0.005M$ $Pb(NO_3)_2$ solution without any precipitation of PbI_2 is $(I=127)$:

(A) $0.09g$
(B) $1.2\times10^{-3}g$
(C) $6\times10^{-4}g$
(D) $1.08\times10^{-5}g$

40. The number of moles of $Cr_2O_7^{2-}$ needed to oxidise 0.136 equivalent of $N_2H_5^+$ through the reaction:

$N_2H_5^+ + Cr_2O_7^{2-} \longrightarrow N_2 + Cr^{3+} + H_2O$ is:

(A) 0.0226
(B) 0.087
(C) 0.136
(D) 0.488

41. 

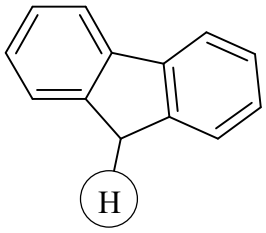
Mono chloro products: (including stereoisomers)

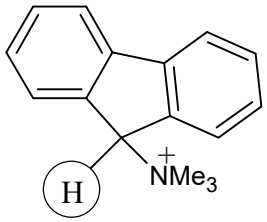
(A) 2
(B) 3
(C) 4
(D) 5

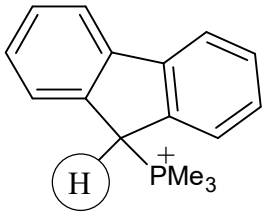
42. The final product obtained from the electrolysis of 50% H_2SO_4 with high current density has:

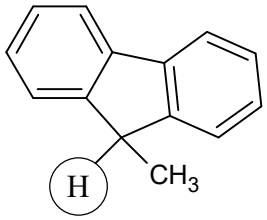
(A) S-O-O-S linkage
(B) S=S linkage
(C) S-S linkage
(D) S-O-S linkage

43. Which of the following circled hydrogen is most acidic?

(A) 

(B) 

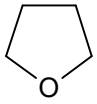
(C) 

(D) 

44. The gradual decrease in atomic radius for lanthanoids elements in not obeyed by:

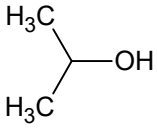
(A) Eu only
(B) Yb only'
(C) Both Eu & Yb
(D) None of these

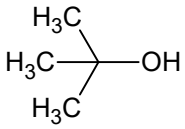
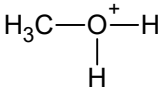
45. Compare the rate of reaction of following species with OH^- :

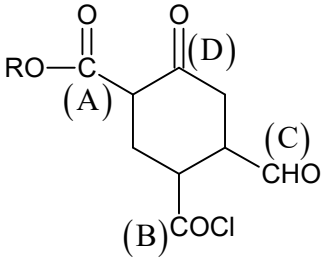
(i)  (ii)  (iii) 

(A) i>ii>iii
(B) i>iii>ii
(C) iii>ii>i
(D) ii>i>iii

46. Which of the following is stronger acid than methanol?

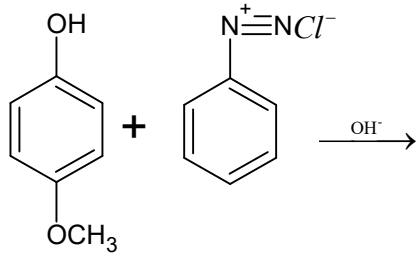
(A) $CH_3 - C \equiv CH$ (B) 

(C)  (D) 

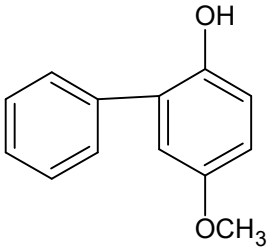
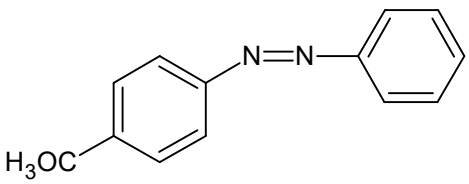
47. 

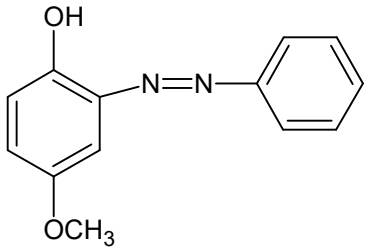
Find the correct order of reactivity of carbonyl groups with $LiAlH_4$:

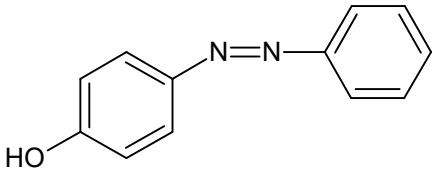
(A) A>B>C>D
(B) B>C>D>A
(C) D>C>B>A
(D) B>D>C>A

48. 

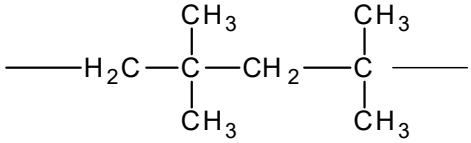
Product of reaction is:

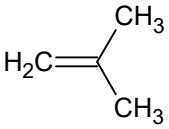
(A)  (B) 

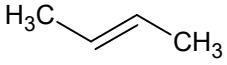
(C) 

(D) 

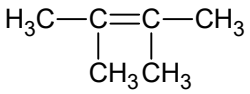
49. The monomer of the polymer

 is:

(A) 

(B) 

(C) $H_3C-CH=CH_2$

(D) 

50. The correct order of matching of complex compound in column I with the properties in column II:

	Column I		Colum II
(A)	$\left[Cr\left(NH_3\right)_6\right]^{3+}$	(P)	Tetrahedral and paramagnetic
(B)	$\left[Co\left(CN\right)_6\right]^{3-}$	(Q)	Octahedral and diamagnetic
(C)	$\left[Ni\left(CN\right)_4\right]^{2-}$	(R)	Octahedral and paramagnetic
(D)	$\left[Ni\left(Cl\right)_4\right]^{2-}$	(S)	Square planar and diamagnetic

(A) A-R, B-Q, C-P, D-S

(B) A-Q, B-R, C-P, D-S

(C) A-R, B-Q, C-S, D-P

(D) A-Q, B-R, C-S, D-P

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

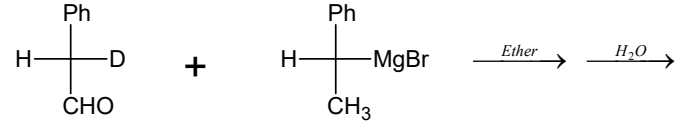
51. In a FCC arrangement W occupies FCC arrangement, X occupies octahedral voids and Y occupies alternate tetrahedral voids, if atoms along two body diagonal are removed then the simplest formula representing the structure is $W_aX_bY_c$ then $\left(a+b-c\right)$ is = ?

52. The number of steps involved in the extraction of Zn from zinc blende is……..
Smelting, froth floatation, distillation, electro-refining, electrolytic reduction, magnetic separation

53. How many of the following statements is/are correct?
(i) Ag sol, As_2S_3, acidic dye stuffs and Haemoglobin are negatively charged colloidal solution.
(ii) Colloidal solution of cellulose nitrate in alcohol is called colloidion.
(iii) Cloud bursts due to mutual discharge of oppositely charged clouds resulting in the coagulation.
(iv) When a freshly prepared precipitate of SnO_2 is peptised by small amount of NaOH, the colloidal particles may be represented as $\left[SnO_2\right]SnO_3^{2-}:2Na^+$, while peptised by small amount of acid, the colloidal particles may be represented as $\left[SnO_2\right]Sn^{4+}$
(v) Tails of comets are visible due to Brownian motion
(vi) Dialysis can separate glucose and protein.

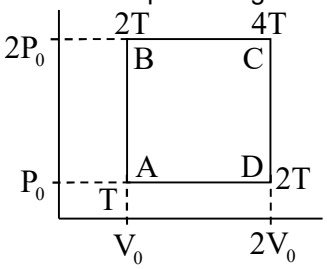
54. How many electrons are present in t_{2g} set of d-orbitals of central metal cation in $\left[Fe\left(H_2O\right)_5\left(NO\right)\right]SO_4$ brown ring complex?

55. The number of stereoisomeric products formed in the following reaction is:



(Optical pure)

56. Find out the percentage efficiency of the following cyclic process for 1-mole mono–atomic gas.



57. A pentapeptide on complete hydrolysis gives Glycine (G), Alanine (A), Leucine (L) and Valine (V). On partial hydrolysis, it never gives Val-Ala. If neither $-NH_2$ or $-COOH$ present at chiral centre, then how many primary structure of pentapeptide are possible?

58. One electrolyzing the solution of $CH_3COONa\left(aq\right)$, the volume ratio of gases formed at anode and cathode is:

59. Equal volumes of 0.2 (M) KI solution and 0.1 (M) HgI_2 solution are mixed. Find the Van't Hoff factor of the resulting solution.

60. In phosgene gas reaction at 400°C, the initial pressure are p_{CO} = 342 mm and $p_{Cl_2} = 352$ mm and the total pressure at equilibrium is 440 mm. $CO + Cl_2 \rightleftharpoons COCl_2$. If percentage dissociation of phosgene at 400°C at 1 atm is x, then find x/10.

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If the coefficient of $b^{12}\, c^6\, d^{15}\, e^{4n}$ in the expansion of $\left(a^7 + b^3 + 2c^2 + d^5 + e^3\right)^{26}$ is non-zero, then its value is

(A) $\frac{|26}{64|20}$

(B) $\frac{|26}{108|16}$

(C) $\frac{|26}{96|16}$

(D) $\frac{|26}{36|20}$

62. The set of values of 'a' for which $ax^2 - (4 - 2a)x - 8 < 0$ for exactly three integral value of x is

(A) $2 \le a < 4$

(B) $1 \le a < 2$

(C) $2 < a < 4$

(D) $1 < a < 2$

63. The number of integers in $2 \le x \le 500$ such that the highest common factor of n and 350 is 1 is

(A) 165

(B) 329

(C) 102

(D) 170

64. The number of values of k for which the expression $\left(2x^2 - 4(k-1)x + 3\right)\left(x^2 + 2kx + k + 1\right)$ is a perfect square is

(A) 0

(B) 1

(C) 2

(D) 4

65. The lines $\frac{x-1}{2} = \frac{y}{-1} = \frac{z}{2}$ & $x - y + z - 2 = 0 = \lambda x + 3z + 5$ are coplanar for λ =

(A) $-\frac{97}{11}$

(B) $-\frac{43}{5}$

(C) $-\frac{73}{9}$

(D) $-\frac{31}{7}$

66. The sum of first n terms (n > 1) of an A.P is 153 and the common difference is 2. If the first term is an integer, the number of possible values of n is
(A) 3
(B) 5
(C) 4
(D) 6

67. Consider the parabola $x^2 + 4y = 0$. Let P = (a, b) be any fixed point inside the parabola and let S be the focus of the parabola. Then the minimum value of SQ + PQ as point Q moves on the parabola is
(A) $|1 - a|$
(B) $|ab| + 1$
(C) $\sqrt{a^2 + b^2}$
(D) $1 - b$

68. Let $f_{n+1}(x) = e^{f_n(x)}$ for all $1 \le n \le 9$ & $f_1(x) = e^x$. Then the first derivative of $f_n(x)$ for n = 10 is
(A) $f_9(x) f_{10}(x)$
(B) $\prod_{r=1}^{9} f_r(x) e^x$
(C) $\prod_{r=1}^{10} e^x f_r(x)$
(D) $\prod_{r=1}^{10} f_r(x)$
where $\prod$ denotes continued product

69. A variable parabola $y^2 = 4ax$, 'a' (where $a \neq -1/4$) being the parameter, meets the curve $y^2 + x - y - 2 = 0$ at two points. The locus of the point of intersection tangents at these points is
(A) $x - 2y - 4 = 0$
(B) $x - 4y + 2 = 0$
(C) $x - 4y - 1 = 0$
(D) $2x - y + 1 = 0$

70. Two vertices of a triangle are (1, 3) and (4, 7). The orthocentre lies on the line x + y = 3. The locus of the third vertex is
(A) $x^2 - 2xy + 2y^2 - 3x - 4y + 36 = 0$
(B) $2x^2 - 4xy + 3y^2 - 4x - y + 42 = 0$
(C) $3x^2 + xy - 4y^2 - 2x + 24y - 40 = 0$
(D) $x^2 - 4xy + 3y^2 - 2x - y + 40 = 0$

71. A seven digit number without repetition and divisible by 9 is to be formed by using 7 digits out of 1, 2, 3, 4, 5, 6, 7, 8, 9. The number of ways in which this can be done is;
(A) 9!
(B) 2 (7!)
(C) 4 (7!)
(D) none of these

72. If the probability of choosing an integer k out of '2m' integers 1, 2, 3, …, 2m is inversely proportional to k^4 $(1 \le k \le 2m)$, then the probability that chosen number is odd, is
(A) > 1
(B) > 1/2
(C) = 1/2
(D) < 1/3

73. The number of ways to place 11 identical balls in three distinct boxes so that any two boxes together will contain more balls than the other one
(A) 63
(B) 21
(C) 15
(D) None of these

74. The equations $(\lambda-1)x+(3\lambda+1)y+2\lambda Z=0,(\lambda-1)x+(4\lambda-2)y+(\lambda+3)z=0,2x+(3\lambda+1)$ $y+3(\lambda-1)z=0.$ gives non-trivial solution for some values of λ, find the ratio x : y : z when λ has the smallest of these values:
(A) 3 : 2 : 1
(B) 3 : 3 : 2
(C) 1 : 3 : 1
(D) 1 : 1 : 1

75. A tangent to the circle $(x-1)^2+y^2=4$ meets the circle $(x-1)^2+y^2=8$ at P and Q. The angle between the tangents at P and Q of the circle $(x-1)^2+y^2=8$ is
(A) 90^0
(B) 60^0
(C) 45^0
(D) None of these

76. The mean of four points in which a circle $x^2+y^2=4$ intersects a hyperbola $xy=4$ is
(A) (3/2, 1)
(B) (1, 3/2)
(C) (0, 0)
(D) None of these

77. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three mutually perpendicular vectors, each of magnitude m units. If $\bar{x}$ satisfies the equation $\vec{a}\times\left\{(\vec{x}-\vec{b})\times\vec{a}\right\}+\vec{b}\times\left\{(\vec{x}-\vec{c})\times\vec{b}\right\}+\vec{c}\times\left\{(\vec{x}-\vec{a})\times\vec{c}\right\}=0$ then $|\vec{x}|$ Is equal to
(A) $\frac{m\sqrt{6}}{2}$
(B) $\frac{m\sqrt{3}}{2}$
(C) $\frac{m\sqrt{3}}{3}$
(D) None of these

78. In a ΔABC, sides a, b, c are in A.P., then $\sin B \sin C \cos^2 A/2$, $\sin A \cos^2 B/2 \sin C$ and $\sin A \sin B \cos^2 C/2$ are in
(A) G.P.
(B) H.P.
(C) A.P.
(D) None of these

79. Two subsets A and B of a set containing n elements are chosen at random. The probability that $A \subseteq B$ is
(A) $\frac{1}{2}$
(B) $\frac{2^n}{n!}$
(C) $\left(\frac{2}{3}\right)^n$
(D) $\left(\frac{3}{4}\right)^n$

80. If the line $x = y = z$ intersect the line $\sin Ax + \sin By + \sin Cz = 2d^2$, $\sin 2Ax + \sin 2By + \sin 2Cz = d^2$ then $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ is equal to (where $A + B + C = \pi$)
(A) $\frac{1}{16}$
(B) $\frac{1}{8}$
(C) $\frac{1}{32}$
(D) $\frac{1}{12}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. The system of equation $2x + 3y - z = 0$, $3x + 2y + kz = 0$, $4x + y + z = 0$ have a set of nonzero integral solution then the least positive value of z is _____

82. The maximum integral value of the function $f(x) = \int_{\pi/2}^{x} (\cos t - 2\sin t)dt$ in the interval $\left[\frac{\pi}{2}, \pi\right]$ is_____

83. The number of points in (–5, 5) at which the function $f(x) = x\sin\left([x]^4 - 5[x]^2 + 4\right)$ is discontinuous is _____

84. The projection of the line $\frac{x}{2}=\frac{y-1}{2}=\frac{z-1}{1}$ on a plane P is $\frac{x}{1}=\frac{y-1}{1}=\frac{z-1}{-1}$. The plane P passes through (k, –2, 0) then k = _____

85. 6 boys, 5 girls and 3 teachers are arranged in a line such that boys are in ascending order girls are in decreasing order, no two teachers are together. The number of such arrangements is 220 K. $^{11}C_5$ then k = _____

86. The differential equation $dy = x\tan y(\sin y - 1)dx$ has solution $y = \sin^{-1}\left(\frac{1}{k + ce^{x^2/2}}\right)$ then $k =$ ___

87. The sum of three positive integers is 20. The probability that they form the sides of a triangle is P then 19P = _____

88. y-axis is a tangent to one ellipse with foci (2, 0) and (6, 4). If the length of the major axis is x then [x] = _____

89. The number of pairs (x, y) of real numbers with $0 < x < \frac{\pi}{2}$ such that $\frac{(\sin x)^{2y}}{(\cos x)^{\frac{y^2}{2}}}+\frac{(\cos x)^{2y}}{(\sin x)^{\frac{y^2}{2}}}=\sin 2x$ is _____

90. Two equal parabolas with vertical axis of symmetry p_1 and p_2 have their vertices at (0, 4) and (6, 0) respectively. If they touch each other at a point, the sum of abscissa and ordinate of point of contact is _____